Problem Statement/Concept Title: HealneX – Seamless Healthcare Innovation

Reimagining healthcare accessibility and efficiency using technology

Theme: Healthcare
Team Name: CodeX
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Institution: Vishwakarma Government Engineering College (VGEC)
State: Gujarat

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Problem – Proposed Solution

Hack for Social Cause

Core Problems

- Limited access in rural & underserved areas
- Disconnected medical records across providers
- High consultation & treatment costs
- Poor continuity of care and follow-up
- Trust gap towards digital consultations

Proposed Solution – HealneX

- Unified digital platform for patients, doctors & admins
- Quick OTP registration & discover doctors easily
- Book appointments & pay securely online
- Teleconsultation via video/voice/chat
- Upload & manage medical records securely
- AI insights, rewards & referral loops for engagement

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Technical Approach

- Proven web & mobile tech; scalable microservices – Python Flask
- Integration with payment & messaging APIs – Stripe & Twilio
- Cloud-ready, modular & multi-region deployment - Render
- AI analytics for personalised recommendations - Transformers Model

Register → Discover Doctor → Book & Pay → Teleconsult → Upload Records → Follow-up

- User Service
- Appointment Service
- Payment Service
- Teleconsult Service
- Record Service
- Notification Service

Live Prototype
Demo Video
Github

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Impact Potential

Hack for Social Cause

Access
Improved access for remote populations

Cost
Reduced healthcare costs

Continuity
Better continuity & outcomes

Trust
Increased trust & transparency

Target Audience & Stakeholders

- Patients (remote & underserved)
- Doctors & clinics
- Administrators & policy makers
- Technology partners & regulators

Expected Measurable Impact

- Higher adoption of teleconsultations & reduced wait times
- Lower out-of-pocket expenses for patients
- Enhanced continuity & adherence to care
- Greater trust & satisfaction in digital health

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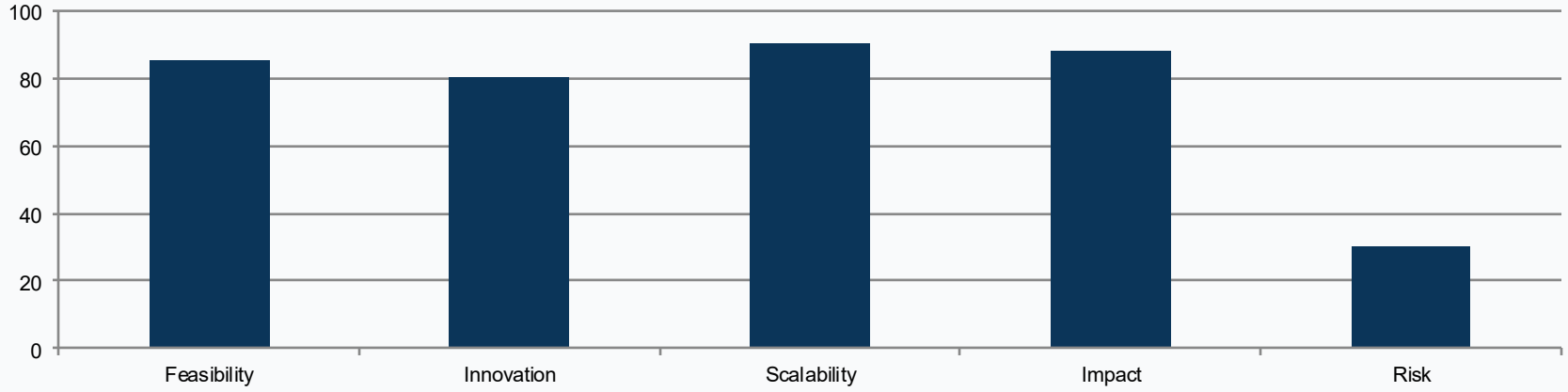
Feasibility & Risk Analysis

Feasibility & Scalability

- Built with proven web & mobile tech
- Integrates payment & messaging APIs
- Modular microservices architecture
- Cloud-ready & multi-region support
- Adaptable to diverse regulations & languages

Risks & Mitigation

- Digital divide & adoption barriers
- Data security & privacy concerns
- Regulatory compliance challenges
- High implementation & maintenance costs
- Mitigation: pilot programmes, robust encryption, partnerships & user education



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Team

Ram

Role: Developer & Backend Lead

Responsibilities:

- Architecture & APIs
- Database & Security
- Deployment & Testing

Nisarg

Role: Designer & Frontend Lead

Responsibilities:

- UI/UX & Frontend
- User Research & Documentation
- Presentation & Demo

Milan

Role: AI/ML Lead

Responsibilities:

- Model Research and Data Study
- Model Training Evaluation and Optimization
- Dataset Preparation

Problem Identification

Recognised healthcare gaps

Prototype Development

Built POC & MVP

Demo & Pilot

Showcased at Hackathon

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References

Hack for Social Cause

- National Digital Health Mission guidelines
https://ndhm.gov.in
- Ayushman Bharat Digital Mission health records framework
https://abdm.gov.in
- Telemedicine Practice Guidelines by Ministry of Health
https://www.nmc.org.in
- WHO Digital Health Guidelines
https://www.who.int/publications/i
- UN Sustainable Development Goals
https://sdgs.un.org/goals
- NITI Aayog reports on digital health in India
https://www.niti.gov.in
- World Bank insights on telemedicine in rural areas
https://www.worldbank.org